Regression and Residuals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Use signed subtraction: 7−10=-3.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Distinguish numeric prediction from classification. Try the recall before opening notes.

Your learning target

I can distinguish numeric **prediction** from classification.

I can calculate signed **residual**s with a stated convention.

I can use **residual** patterns to question a model.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

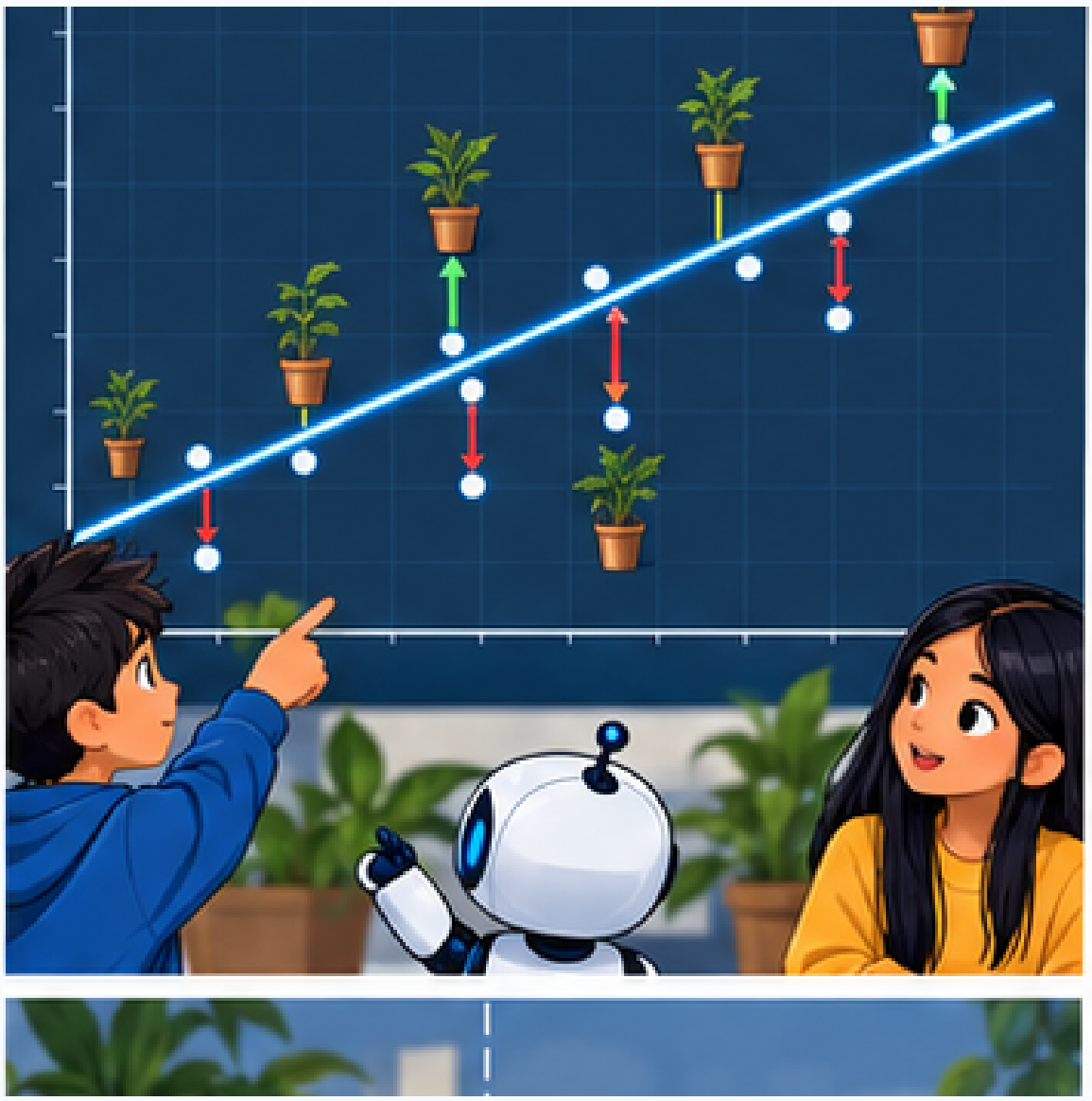
Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Regression predicts a number; a **residual** records signed error and reveals direction as well as size.



Your target

LEARNING TARGET Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual−predicted, calculate **residual** and absolute error. Interpret the sign.

Builder name and experiment date:

Readiness for the connected explanation

What does a class label predict?

This week's end goal

By the end, I can explain **regression** and **residual**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

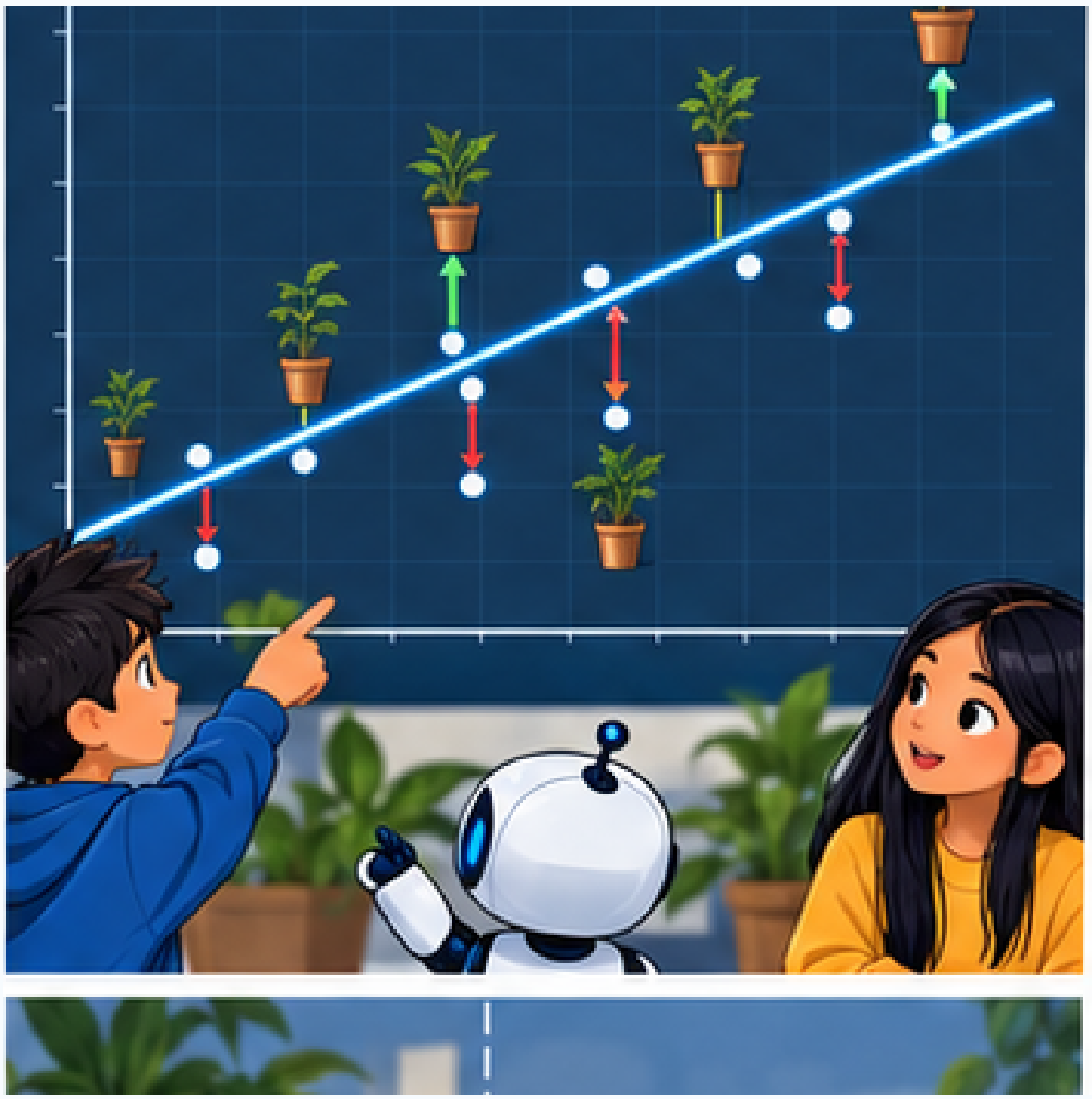
Notice the evidence

> Investigate the supplied Regression and Residuals case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **regression** and **residual**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A school garden example predicts plant height in cm from recorded conditions. The target is numeric, so a **prediction** can be too high or too low by different amounts. Define **residual** as observed minus predicted.

Keep units and the sign convention visible; some software uses the opposite error convention.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **regression** and **residual**s.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Regression and Residuals. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.

IN

INPUT

Plot predictions and calculate positive and negative residuals.

ML

MODEL

Regression predicts a number; a residual records signed error and reveals direction as well as size.

OUT

EVIDENCE

Plot predictions and calculate positive and negative residuals.

Question to explain

QUESTION Plot **prediction**s and calculate positive and negative **residual**s.

MODEL IDEA **Regression** predicts a number; a **residual** records signed error and reveals direction as well as size.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Actual plant height is 30 cm; **prediction** is 27 cm. **Residual** = actual – predicted = 30 – 27 = +3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Observed heights [4, 7, 9] cm and **prediction**s [5, 6, 9] cm give **residual**s [-1, 1, 0] cm. The first **prediction** is too high, the second too low and the third exact. Mean signed **residual** is zero, yet two **prediction**s are wrong.

Opposite errors can cancel.

```
observed = [4, 7, 9]
predicted = [5, 6, 9]
residuals = [y - p for y, p in zip(observed, predicted)]
print(residuals)
print(sum(residuals) / len(residuals))
# Output: [-1, 1, 0] and 0.0.
# Opposite signed errors can cancel.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

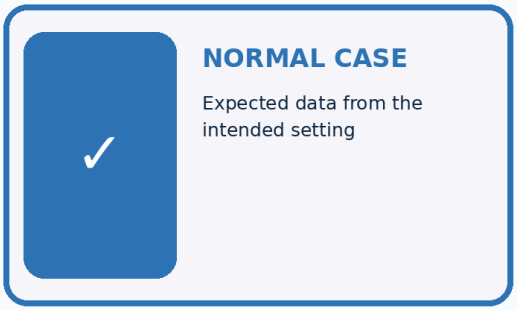
Compare the cases

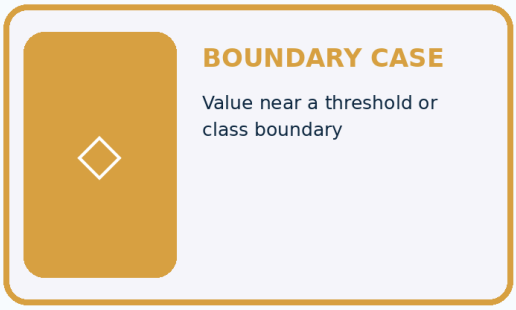
> Compare the supplied cases for Regression and Residuals. Identify the detail that changes the conclusion.

EVALUATION DESIGN

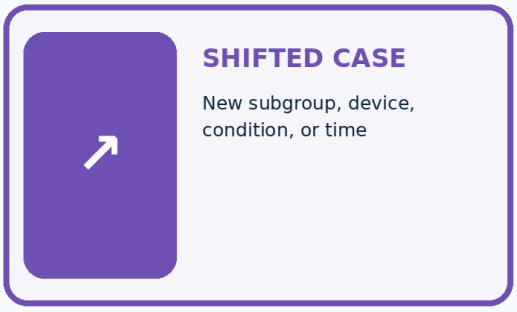
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Raise all **prediction**s by1 cm.

Predicted result

Prediction: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

Why the result follows

Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Regression and Residuals to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

REGRESSION

prediction of a numerical target

2

PREDICTION

model-estimated value

3

RESIDUAL

actual value minus predicted value

4

RESIDUAL PLOT

errors displayed against predictions or inputs

Term	Precise meaning	My ML example
regression	prediction of a numerical target	
prediction	model-estimated value	
residual	actual value minus predicted value	
residual plot	errors displayed against predictions or inputs	

MISCONCEPTION TO REPAIR Positive and negative **residual**s cancel, so their average proves accuracy.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Regression	Prediction of a numeric target
Residual	Observed target minus predicted target in this course
Underprediction	A prediction below the observed value
Overprediction	A prediction above the observed value

STUDENT LESSON · 6 OF 18

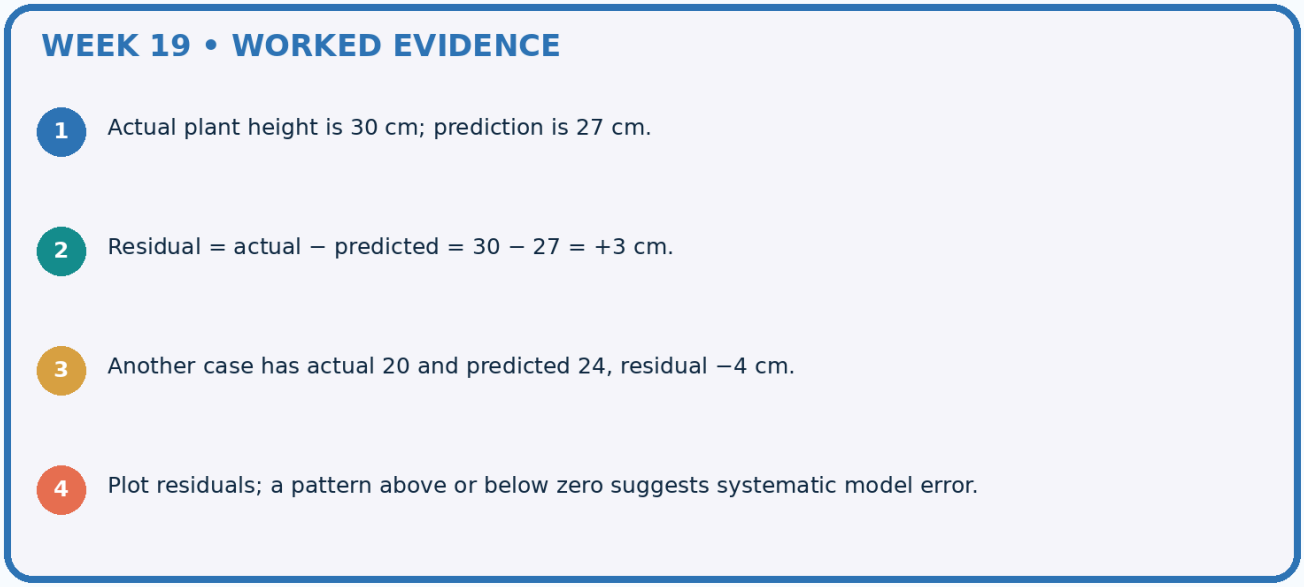
Follow a worked example

> Follow the worked trace for Regression and Residuals, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



Step 1

STEP 1 Actual plant height is 30 cm; **prediction** is 27 cm.

Step 2

STEP 2 **Residual** = actual – predicted = 30 – 27 = +3 cm.

Step 3

STEP 3 Another case has actual 20 and predicted 24, **residual** –4 cm.

Step 4

STEP 4 Plot **residual**s; a pattern above or below zero suggests systematic model error.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Regression and residuals at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Observed 4 predicted 5	Residual minus 1	Overprediction
Observed 7 predicted 6	Residual plus 1	Underprediction
Observed 9 predicted 9	Residual zero	Exact on this record

To inspect a residual plot, place a feature or prediction on the horizontal axis and residual on the vertical axis. A trend can suggest a missed relationship. It does not by itself identify the cause or validate a new model. Use development data for analysis and preserve an independent final test.

Why does mean residual zero not imply perfect predictions? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Regression and Residuals

Use **residual** signs to say whether each plant-height estimate is too high or low.

The facts we will use

Supplied facts

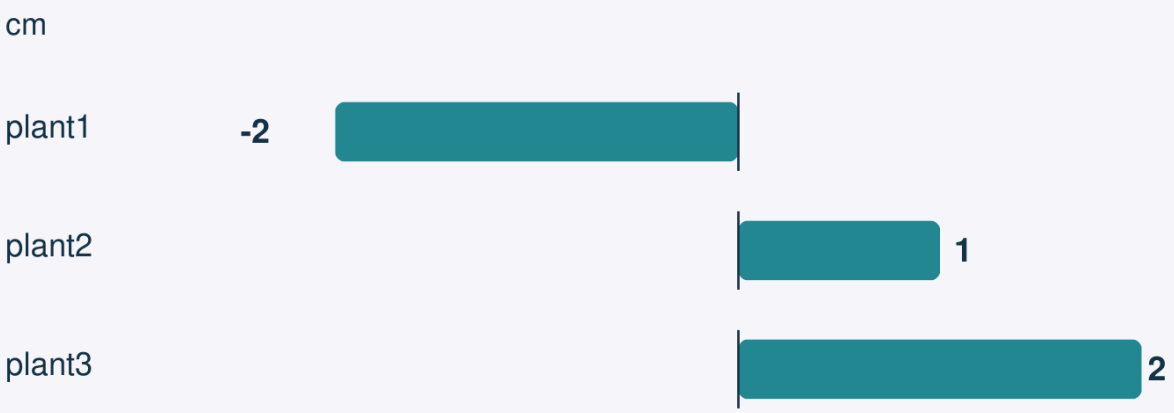
Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/19

Observed minus predicted height



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/19

Regression and Residuals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Subtract matched pairs

Residuals are5-7=-2, 8-7=1, 12-10=2 cm.

2. Read directions

First prediction is too high by2; second too low by1; third too low by2.

3. Avoid cancellation

Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

CHANGE ONE THING

Raise all predictions by1 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Subtract matched pairs	Residuals are5-7=-2, 8-7=1, 12-10=2 cm.
2. Read directions	First prediction is too high by2; second too low by1; third too low by2.
3. Avoid cancellation	Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Always declare the **residual** sign convention; some software uses predicted minus

observed.

Keep the stated limit

Read the labelled picture: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Raise all **prediction**s by1 cm. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Regression and Residuals claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Positive and negative residuals cancel, so their average proves accuracy. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Positive and negative **residual**s cancel, so their average proves accuracy.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A **residual** of zero is an agreement with the measured target, not proof that the measurement is perfect. Do not compare errors measured in cm directly with errors measured in metres.

Keep the same row pairing, unit and sign convention throughout.

A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.

STUDENT LESSON · 8 OF 18

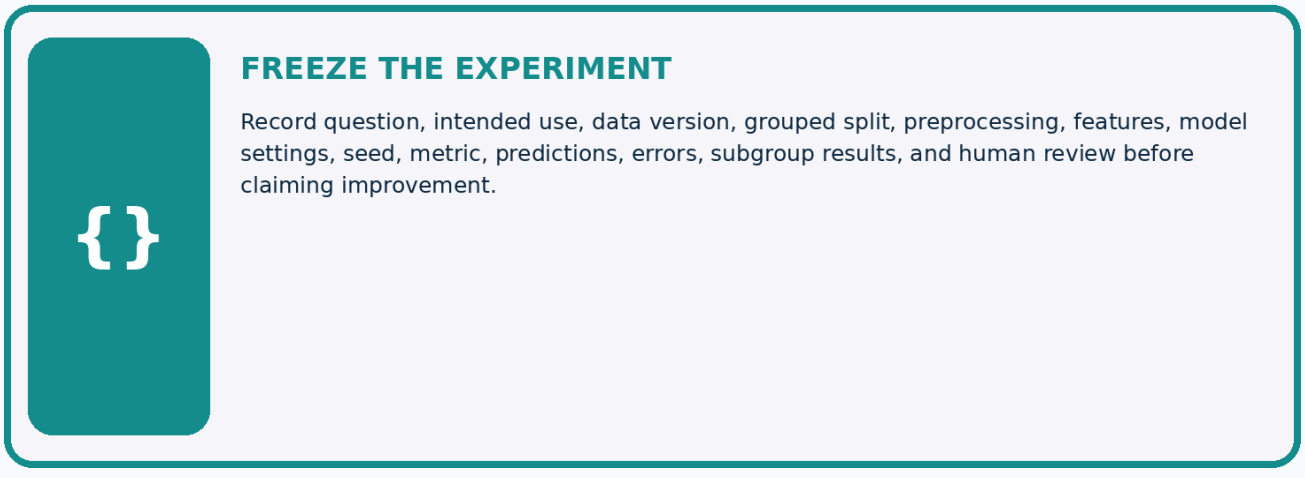
Practise together

Practise Regression and Residuals with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If **prediction**s for early, middle and late growth have **residual**s -2, 0, 3 cm, the model moves from overpredicting to underpredicting across those records.

Investigate whether a straight-line assumption or a missing condition could explain it. Check measurement consistency before adding complexity.

Practise with less help: Explain cancellation

Use workbook W2 for the connected example "**Regression** and **residual**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Regression** and **residual**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Regression and Residuals. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Residual Walk: calculate signed errors, place them on a plot, and identify bias, outliers, or changing spread.

1. Write the hypothesis and expected **evidence** before running.

2. **Residual** Walk: calculate signed errors, place them on a plot, and identify bias, outliers, or changing spread.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

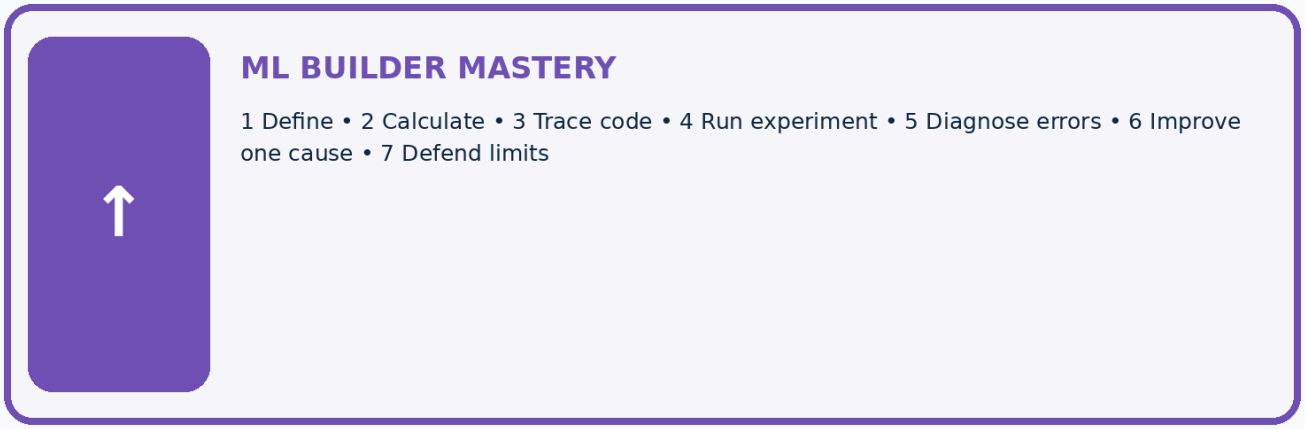
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Regression and Residuals. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **regression**, **prediction**, **residual**, **residual plot**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Positive and negative **residual**s cancel, so their average proves accuracy.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Use a miniature growth-prediction side experiment to distinguish regression evidence from the main classifier metrics.

THIS WEEK'S CHECKPOINT Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

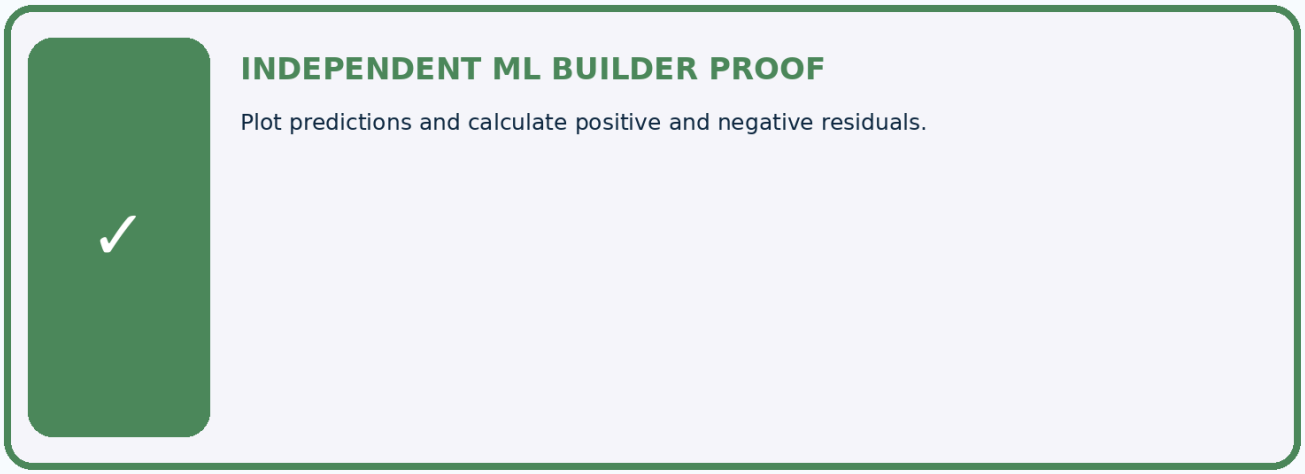
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Plot **prediction**s and calculate positive and negative **residual**s.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Mean, Median, and Baselines.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Regression and Residuals case for supported practice. Repeated examples do not become fresh assessments.

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

The question

Calculate and interpret.

Worked explanation

+3 cm means **underprediction** by 3 cm. State this sign convention consistently.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Residual is actual minus predicted

Evidence or stage	Value or result
Prediction	9
Actual	12

Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

Calculate and interpret.

+3 cm means **underprediction** by 3 cm. State this sign convention consistently.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Define **residual** as actual minus predicted. Actual height is 14 cm and **prediction** is 17 cm. Calculate **residual** and interpret its sign.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual–predicted, calculate **residual** and absolute error. Interpret the sign.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

Use this map to navigate Regression and Residuals, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Use **residual** signs to say whether each plant-height estimate is too high or low.

By the end, I can	How I will show it
Distinguish numeric prediction from classification	Explain the deciding step, show a trace or test, and state one limit.
Calculate signed residuals with a stated convention	Explain the deciding step, show a trace or test, and state one limit.
Use residual patterns to question a model	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Regression and Residuals without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish numeric **prediction** from classification

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use **residual** signs to say whether each plant-height estimate is too high or low.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

1. Read directions

2. Avoid cancellation

3. Subtract matched pairs

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Avoid cancellation Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Read the picture in order.

Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Raise all **prediction**s by1 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Each **residual** decreases by1 under the observed-minus-predicted convention.

Repair a missing building block

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

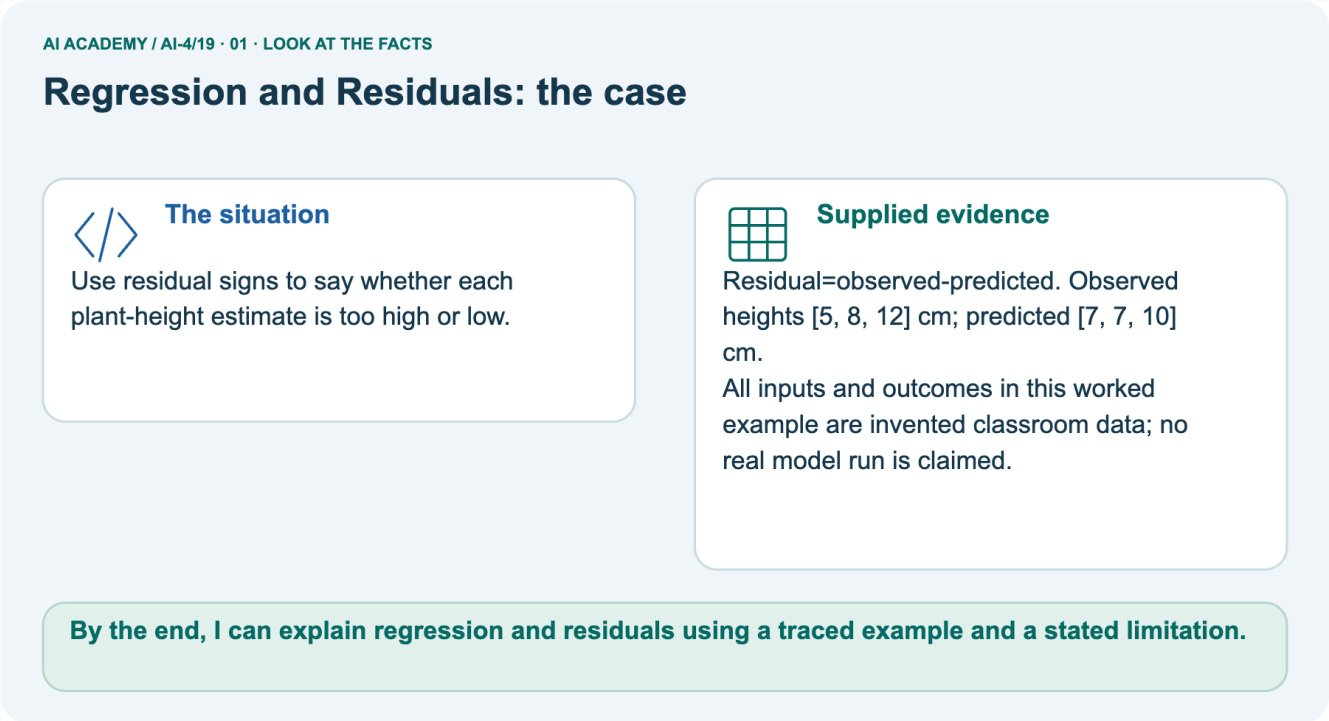
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Regression and Residuals graphic using its labels, deciding evidence and relevant limitation.

01. Regression and Residuals: the case



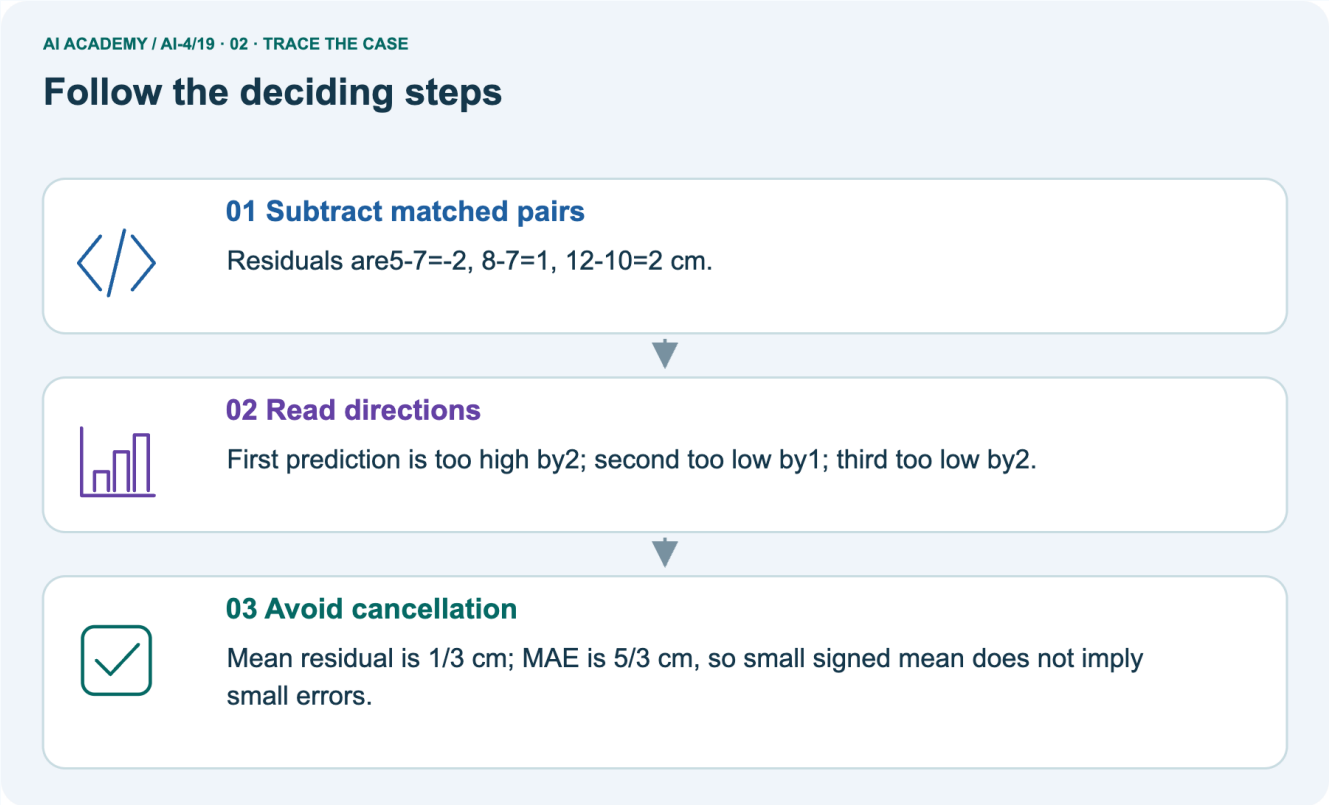
Follow the labelled picture with the matching explanation. Open the vector diagram

Use **residual** signs to say whether each plant-height estimate is too high or low. Supplied facts: **Residual**=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Subtract matched pairs: **Residual**s are5-7=-2, 8-7=1, 12-10=2 cm. → Read directions: First **prediction** is too high by2; second too low by1; third too low by2. → Avoid cancellation: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Observed minus predicted height

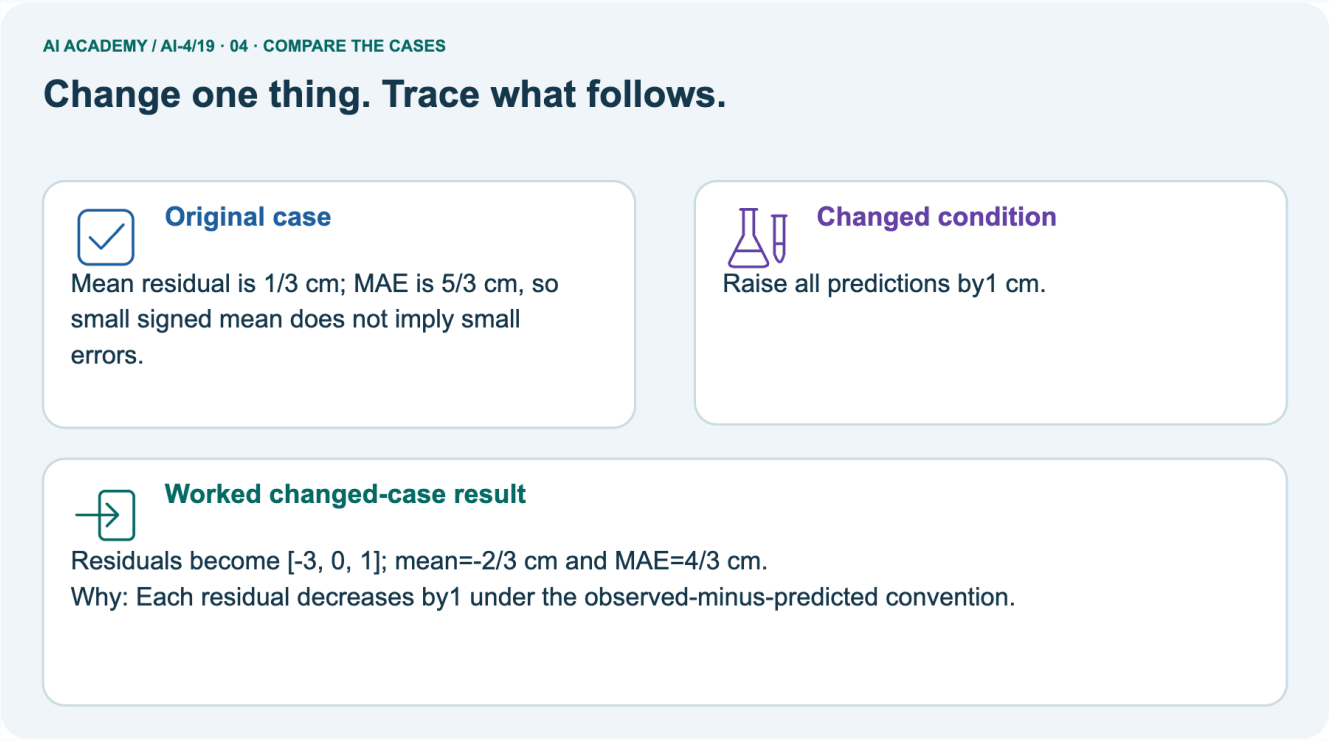


Follow the labelled picture with the matching explanation. Open the vector diagram

Observed minus predicted height. Units: cm. plant1: -2. plant2: 1. plant3: 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



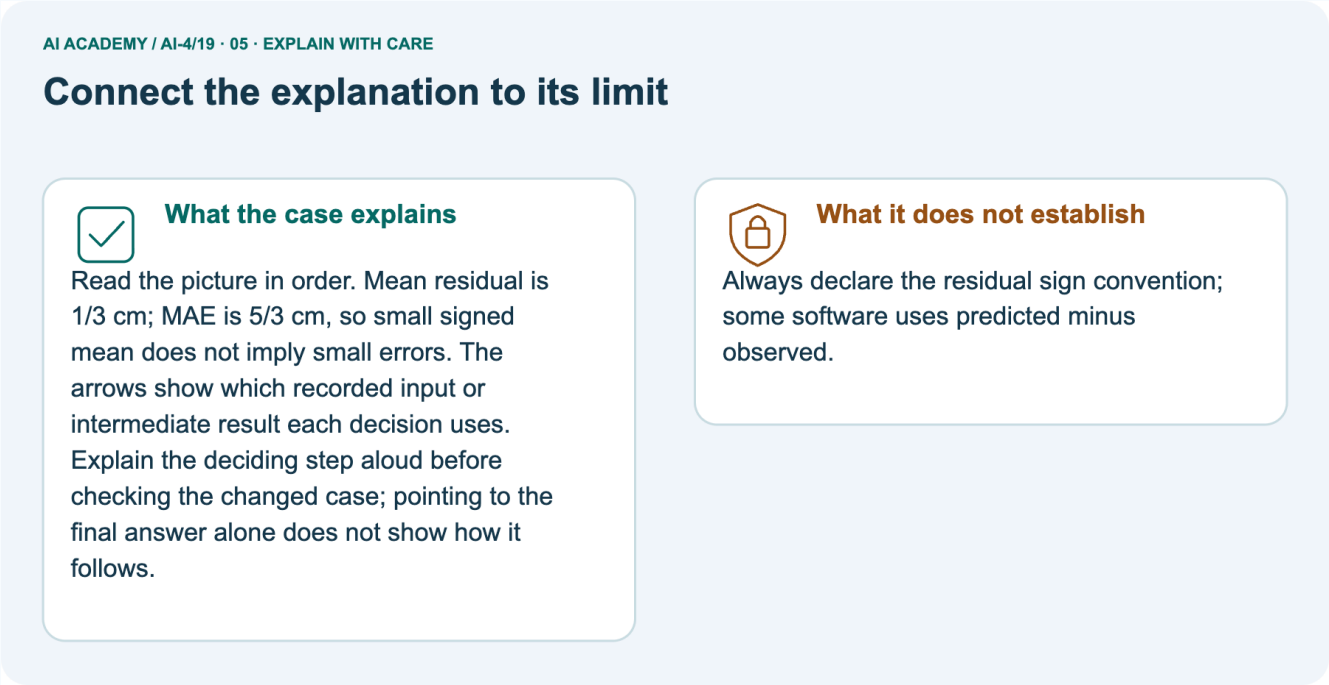
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Changed condition: Raise all **prediction**s by1 cm.

Expected result: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

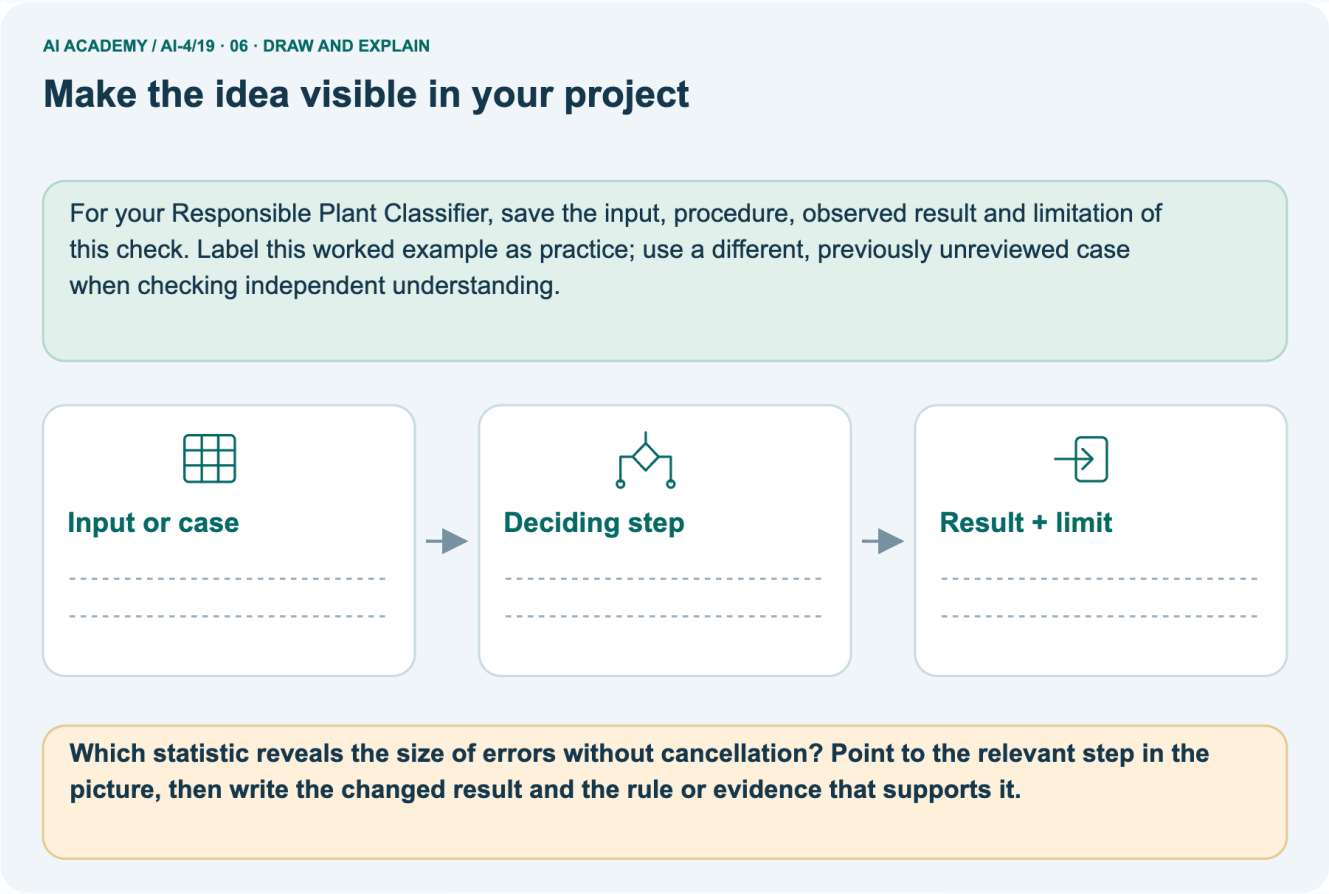
Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Always declare the **residual** sign convention; some software uses predicted minus observed.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.